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Forest Conservation – Too Much or Too Little? A Political Economy Model

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Abstract. This paper studies the formation of forest policy when the government is influenced by an environmental lobby and an industrial lobby representing a non-competitive wood processing industry. Government decides on forest conservation by way of restricting timber harvesting. Lobbying is modelled as a common agency game with differences in the efficiency of lobbying. A comparison of the political equilibria shows that an exporting forest industry faces a stricter conservation requirement than a forest industry whose production is destined for domestic markets. If the industrial lobby is more efficient than the environmental lobby, conservation is insufficient from the social point of view. However, conservation may be insufficient even if the environmental lobby is more efficient in lobbying than the industrial lobby. This is because the lobbying effort of the environmental lobby also benefits consumers that remain politically passive.

Key words: amenity valuation, common agency, forest policy, lobbying, market power

JEL classifications: D72, Q23

1. Introduction

Forest resources are used to produce timber, but they also provide many kinds of amenities. Timber and related wood products are private goods whereas other services provided by the forests are to a large extent public. If the forest owner is not compensated for the external benefits created by the forestry, the level of socially valuable public services of the forests may remain too low.[1]

Government forest policies aim to create a balance between these different uses of forest resources. Typically people are not affected in the same way by the government's decisions, and therefore, the trade-off between different uses of forest resources creates political tensions. Groups that often participate in the public debate on the 'correct' objective of forest policy include environmentalists, the wood processing industry, and the forest owners.[2]

Despite the debates concerning government forest policies, the formation of forest policy has not been previously analyzed in the political economy literature. How the aims of different groups within the society are reflected in the formation of forest policy is the topic of this paper. The main questions of interest are under what conditions and how the influence of special interests can be expected to cause observed policies to be distorted from socially optimal policies. In order to study these questions, we consider a specific policy issue, the conservation of forests by way of restricting timber harvesting in areas that are particularly valuable for non-timber services of forests. If conservation makes harvesting more costly for the forest owner, it may also increase the cost of timber for the wood processing industry. At the same time, conservation guarantees that part of the forest resources is used for recreational and environmental purposes. For simplicity, we consider a situation where the wood processing industry owns the forests subject to conservation. Clearly, in this situation the forest industry and environmental groups have different preferences over the government's decisions and may seek to influence policy making in order to enhance their welfare.

We will analyze the situation illustrated above using the common agency model introduced by Bernheim and Whinston (1986). In the political context this model was first used by Grossman and Helpman (1994) to analyze endogenous trade policy, and it has since then become a standard model to analyze government policies under interest group influence.[3]

One of the characteristics of wood processing industries is often a concentrated market structure.[4] Furthermore, in countries where the wood processing industry is important for the whole economy, changes in the price of timber may have repercussions for several sectors of the economy. As a result, forest conservation decisions affect not only the welfare of the groups that actively participate in the debate on forest conservation, but the welfare of all consumers. Finally, an industrial lobby representing a concentrated wood processing industry and an environmental lobby typically have distinct methods of lobbying and different channels of influence. We aim to capture the features mentioned above in the following way. First, we will consider an industrial lobby that represents a non-competitive industry, which we for simplicity model as a monopoly. Second, imports will not be a perfect substitute for domestic production of the wood processing industry. These features of the model imply that forest conservation influences the welfare of all consumers through changes in consumer price. They therefore enable us to study how the costs of conservation are distributed between producers and consumers and how the market power of the wood processing industry influences the political determination of forest conservation. In order to isolate the welfare-effects of forest conservation through the consumer price, we study policy design in two different situations, in a situation where the wood processing industry exports its final product and a situation where its production is destined for domestic markets. Third, we allow for asymmetry between the two lobbies in that they need not be equally efficient in lobbying. This implies that even if the lobbies were to use the same amount of total resources in order to influence policy making, one of the lobbies may be more successful in its effort.

Previous applications of the common agency model to environmental policy design concentrate on environmental and trade policies in small open economies in a setting where firms are competitive and imports are a perfect substitute to domestic production. Fredriksson (1997) considers a small open economy with a non-polluting and a polluting production sector. The government uses a production tax to regulate pollution, and the owners of a fixed factor used in the polluting sector and an environmental lobby seek to influence the policy choices. Aidt (1998) considers a small open economy with several production sectors some of which are represented by a lobby. All sectors use polluting raw materials in production. He shows that while a benevolent social planner would use only a tax on the raw material use, in the political equilibrium the government uses a tax on the raw material use as well as output taxes and subsidies. Production sectors that are represented by a lobby receive an output subsidy while the unorganized production sectors pay a tax. Also in Schleich (1999) some of the several production sectors are represented by a lobby. Only one sector is polluting, but the government may use trade and production policies in all sectors in order to correct the externality and satisfy the industry lobbies. Schleich shows that in the case of production externality the government will use only production policies. The production tax for the polluting sector will be lower than the Pigouvian tax rate, if it is organized, and higher, if it remains unorganized. Moreover, the organized non-polluting sectors get a production subsidy while the unorganized sectors pay a production tax.

The rest of the paper is organized as follows. In section 2, we describe the economy and characterize the socially optimal conservation policy. In section 3, we present the political process. In section 4, we analyze the determinants of the political equilibrium and study how it compares to the social optimum. Section 5 offers some concluding remarks.

2. The Economy

2.1. PRODUCTION

The economy has two production sectors. The first sector is characterized by competitive firms producing a tradable numeraire good, z, with linear technology and labor, n, as the only factor of production. All consumers own one unit of labor which they supply to either of the production sectors. Consumers earn wage, w, in both sectors. We assume that the supply of labor is large enough for the numeraire good to be produced.

The other production sector consists of a monopoly that produces a wood product, y, with timber, k, and labor with constant returns to scale technology, $y = f(k, n)$. The monopoly may use timber from its own forests or import timber for production from abroad. By exporting timber the monopoly earns the world market price for round wood, $\overline{q}$, which thus constitutes the opportunity cost of using timber in production. The cost of importing round wood for production is $\overline{q} + \tau$, where τ represents transportation costs.

As our focus is on the political process, we want to model the economy in the simplest possible way. Therefore, we abstract from the intertemporal aspects of forestry and consider steady-state annual yields. In each period, the monopoly harvests this yield, or part of it, depending on the (deterministic) world market price for round wood.

Conservation reduces forest areas available for commercial harvesting, creating a constraint on cuttings. Conservation decisions therefore include both a decision on what the level of conservation should be and a decision on which particular areas are chosen for conservation. Here, we will concentrate on the decision of the conservation level. The level of conservation is defined to be

$$m = K - \overline{k},$$

where K is the steady-state annual yield and $\overline{k}$ is the amount of timber that may be harvested. Conservation level $m = K$ would then imply a complete preservation policy and conservation level $m = 0$ would imply a market solution.

We will analyze two different situations: domestic consumption of the wood product and exported wood product.[5] The resource constraint for the economy is

$$Y = C + X - M,$$

where Y is national income, C consumption, and X and M the value of exports and imports, respectively. Imports consist of imported numeraire good, z_M, and round wood, k_M, i.e.,

$$M = z_M + k_M\left(\overline{q} + \tau\right).$$

When the wood product is exported, exports consist of the wood product, y, the numeraire good, z_X, and round wood, k_X, i.e.,

$$X = p\left(y\right) y + z_X + k_X\overline{q},$$

where $p\left(y\right)$ is the demand schedule faced by the wood processing industry abroad.

If round wood for production were imported, changes in conservation policy would have no effect on production costs nor the consumer price. We will therefore focus on a situation where $k_M = 0$. In that case, $k_X = \overline{k} - k$, where k is the amount of timber used in own production. National income without the benefits from forest conservation can then be written as

$$Y = z + p\left(y\right) y + \left(\overline{k} - k\right)\overline{q}$$

consisting of domestic production of the numeraire good, the value of the wood product, and the exported round wood.

Given that the monopoly does not import round wood for production, it maximizes profits given by

$$\pi = p(y) f\left(k, n\right) + \left(\overline{k} - k\right)\overline{q} - wn \tag{1}$$

subject to the constraint that $\overline{k} \geq k$.

The total cost of timber use consists of the opportunity cost of timber and the shadow price of harvesting. This shadow price is strictly positive when all timber harvested is used for own production, that is, when $\overline{k} = k$. In that case, tightening the harvesting constraint increases the cost of timber use.[6] Let $q(m)$ denote the total cost of timber use as a function of the conservation level. From the definition of the conservation level it then follows that $q_m(m) > 0$.[7] Thus the marginal effect of conservation on the unit cost of production is

$$c_m(q(m), w) = c_q(q(m), w)\, q_m(m) > 0.$$

The above discussion implies that when analyzing the incentives to use resources in lobbying, of particular interest is the situation where the constraint on harvesting is such that all timber harvested is used for own production and conservation is not extensive enough to induce imports. In this situation changes in conservation policy influence the cost of production and the price of consumption.

2.2. CONSUMERS

The economy consists of three kinds of consumers. The overall size of the population is normalized to one. Fraction α of the population owns the timber using monopoly and earns monopoly profits. In the political process, consumers belonging to this group form the industrial lobby. Fraction β of the consumers belongs to the environmental lobby.[8] These groups are assumed to be disjoint. Fraction $(1 - \alpha - \beta)$ of the consumers belongs to neither of the above mentioned groups and remains passive in the political process.

Consumers belonging to lobby i contribute to their group's objective by providing their share of the total resources, $T_i(m)$, used by the lobby. In this setting, it is natural to think of the environmental lobby as a legitimate, national environmental organization that participates to the formation of government policies. From the point of view of an individual that belongs to the environmental organization, the contribution could be interpreted as a membership fee.[9] Net income of consumers in different groups are then

$$\begin{aligned} I_\alpha &= w + \frac{1}{\alpha}\pi(p, q(m), w) - \frac{1}{\alpha}T_\alpha(m), \\ I_\beta &= w - \frac{1}{\beta}T_\beta(m), \\ I_{1-\alpha-\beta} &= w. \end{aligned}$$

When there exists a domestic market for the wood product, the utility maximizing demands for the wood product and the numeraire good are derived as solutions to the following optimization problems:

$$\max_{z,y} \{z + u(y) + g(m)\} \text{ s.t. } I_i = z + py \text{ for } i = \alpha, \beta, 1 - \alpha - \beta,$$

where $u(y)$ is utility from consumption of the wood product, $g(m)$ is utility from conservation, and p is the price of the wood product.[10] All members of the society benefit equally from conservation. This assumption highlights the idea that the most important non-timber services of forests are pure public goods, like forests as a stock for carbon and biodiversity.[11]

Since the utility function is quasi-linear, the demand of the wood product depends only on its own price and can be expressed as $y = d(p)$. The demand of the numeraire good is then

$$z_i(p) = I_i - pd(p) \text{ for } i = \alpha, \beta, 1-\alpha-\beta,$$

where I_i is the net income of a consumer in group i. Given $d(p)$ and $z_i(p)$, the policy preferences of the consumers in different groups can be represented by the following indirect utility functions:

$$\begin{aligned} V_\alpha(m) &= w + cs(m) + g(m) + \frac{1}{\alpha}\pi(p, q(m), w) - \frac{1}{\alpha}T_\alpha(m) \\ &= \frac{1}{\alpha}\left[\phi_\alpha(m) - T_\alpha(m)\right], \\ V_\beta(m) &= w + cs(m) + g(m) - \frac{1}{\beta}T_\beta(m) \\ &= \frac{1}{\beta}\left[\phi_\beta(m) - T_\beta(m)\right], \\ V_{1-\alpha-\beta}(m) &= w + cs(m) + g(m) \\ &= \frac{1}{1-\alpha-\beta}\phi_{1-\alpha-\beta}(m), \end{aligned} \tag{2}$$

where $cs(m) = u(d(p(m))) - p(m)\,d(p(m))$, and the gross welfare of group i is $\phi_i(m)$.

2.3. SOCIALLY OPTIMAL CONSERVATION POLICY

We now study how conservation influences welfare of the different groups in the economy. After that we characterize the socially optimal conservation policy. Aggregate welfare is represented by a utilitarian welfare function. We define two different welfare measures. Aggregate gross welfare consists of consumer surplus, monopoly profits, and utility from conservation. Aggregate net welfare in turn refers to aggregate welfare net of lobbying costs. When the wood product is consumed domestically, the latter is given by $W^d(m) = \phi^d(m) - T_\alpha^d(m) - T_\beta^d(m)$, where

$$\begin{aligned} \phi^d(m) &= \phi_\alpha^d(m) + \phi_\beta^d(m) + \phi_{1-\alpha-\beta}^d(m) \\ &= w + cs(m) + g(m) + \pi(p, q(m), w) \end{aligned} \tag{3}$$

and superscript d refers to domestic consumption.

The effects of a change in conservation on the gross welfare of different consumer groups are given by

$$\frac{\partial \phi_{\alpha}^{d}(m)}{\partial m} = \pi_m\left(p, q(m), w\right) + \alpha\left[cs_m(m) + g'(m)\right], \tag{4}$$

$$\frac{\partial \phi_{\beta}^{d}(m)}{\partial m} = \beta\left[cs_m(m) + g'(m)\right], \tag{5}$$

and

$$\frac{\partial \phi_{1-\alpha-\beta}^{d}(m)}{\partial m} = (1-\alpha-\beta)\left[cs_m(m) + g'(m)\right], \tag{6}$$

where

$$\begin{aligned} cs_m(m) &= -u''\left(y\left(m\right)\right) y\left(m\right) y_m\left(m\right) < 0, \\ \pi_m\left(p, q(m), w\right) &= -y\left(m\right) c_m\left(q\left(m\right), w\right) < 0, \end{aligned}$$

and $y\left(m\right)$ denotes the profit maximizing output level as a function of the conservation level.

A stricter conservation policy reduces consumer surplus as it increases the price of consumption. It also reduces the maximum profits of the wood processing industry. A stricter conservation policy also directly benefits all consumers, and hence the optimal conservation level is strictly positive. Under the assumption that the second-order condition for welfare maximization is satisfied, the socially optimal conservation level, m^{d*}, is determined by

$$\frac{\partial \phi^{d}(m^{d*})}{\partial m} = \frac{\partial \phi_{\alpha}^{d}(m^{d*})}{\partial m} + \frac{\partial \phi_{\beta}^{d}(m^{d*})}{\partial m} + \frac{\partial \phi_{1-\alpha-\beta}^{d}(m^{d*})}{\partial m} = 0. \tag{7}$$

In the social optimum, the marginal benefit of increased conservation equals the cost imposed on consumers and monopoly owners in the form of higher consumption price and lower profits.

When the final product of the wood processing industry is exported, aggregate net welfare is $W^{e}\left(m\right) = \phi^{e}(m) - T_{\alpha}^{e}(m) - T_{\beta}^{e}(m)$, where

$$\begin{aligned} \phi^{e}\left(m\right) &= \phi_{\alpha}^{e}(m) + \phi_{\beta}^{e}(m) + \phi_{1-\alpha-\beta}^{e}(m) \\ &= w + g\left(m\right) + \pi\left(p, q(m), w\right) \end{aligned} \tag{8}$$

and superscript e refers to exported wood product. Now both the environmentalists and the passive consumers always benefit from an increased conservation requirement. In the social optimum

$$\frac{\partial \phi^{e}(m^{e*})}{\partial m} = \frac{\partial \phi_{\alpha}^{e}(m^{e*})}{\partial m} + \frac{\partial \phi_{\beta}^{e}(m^{e*})}{\partial m} + \frac{\partial \phi_{1-\alpha-\beta}^{e}(m^{e*})}{\partial m} = 0. \tag{9}$$

In both cases, in the social optimum, welfare of the members of the environmental group is increasing in m and welfare of the monopoly owners is decreasing in m. This conflict of interests reflects the nature of forests as a joint production of private goods and amenity values.

3. The Political Process

We now turn to the determination of the conservation policy when environmental and industrial lobbies influence the decisions of the government. The government has two distinct objectives. It takes into account the aggregate welfare but may also be willing to distort policy in order to receive benefits from the lobbies. The government therefore maximizes a weighted sum of aggregate welfare and contributions from the two lobbies

$$G(m; C_\alpha^j, C_\beta^j) = a\phi^j(m) + C_\alpha^j(m) + C_\beta^j(m) \text{ for } j = d, e, \tag{10}$$

where parameter a is the weight that the government attaches to aggregate welfare of the consumers and $C_i^j(m)$ denotes contributions of lobby i contingent on the policy chosen by the government. The relative weight of aggregate welfare could be interpreted as a measure of the transparency of the political system.[12]

Restricting harvesting is an instrument that allows for conserving certain forest areas that are particularly valuable for non-timber services. Harvesting taxes, for instance, also typically influence the quantity harvested, but they cannot be targeted to preserve a given area. In addition, using a tax as an instrument to regulate harvesting would change the nature of the political process. Taxation would generate revenue to be distributed back to the consumers. The existence of this revenue would create an additional motive for lobbying. In fact, many of the distortions created by lobbying in the previous literature on environmental regulation with competitive industries stem from this additional motive.[13]

The contributions of the two lobbies represent the monetary equivalent of all the resources used to influence the political process. When considering the influence of two very different kind of lobbies, it is natural to allow for differences in the efficiency of lobbying. We follow Laffont and Tirole (1993, Ch. 11) in assuming that using resources in order to influence the government is costly in the sense that a contribution of one dollar costs $(1 + \lambda_i)$ dollars for lobby i. Therefore, total resources used by lobby i are

$$T_i^j(m) = (1 + \lambda_i)\, C_i^j(m) \text{ for } j = d, e.$$

Parameter λ_i is non-negative and reflects the efficiency of lobbying. When $\lambda_\alpha = \lambda_\beta$, lobbies are equally efficient. In general, λ_i depends on how the attractiveness of lobbying is influenced, for instance, by the tax code and direct regulations. In addition, λ_i is likely to be high for groups with high fund raising costs or high administrative costs, which may arise because of large and dispersed membership. Differences in the cost of lobbying could also arise because the preferences of the government are more in line with one lobby than the other.[14]

Timing of the political game is as follows. Lobbies first choose simultaneously their contribution schedules which determine their contribution for all possible policy decisions of the government. The government then chooses the conservation level that maximizes its payoff. Lemma 2 in Bernheim and Whinston (1986) shows that there exists a set of subgame perfect equilibria in this game. It can be readily extended to the case where lobbying is costly.

Lemma 1 *When there are strictly positive lobbying costs, policy m^j and contribution schedules $C_\alpha^j(m)$ and $C_\beta^j(m)$ form a subgame perfect equilibrium if and only if*

1) $C_i^j(m) \geq 0$ *for* $i = \alpha, \beta$

2) $m^j \in \arg\max \left\{ G(m; C_\alpha^j, C_\beta^j) \right\}$,

3) $m^j \in \arg\max \left\{ G(m; C_\alpha^j, C_\beta^j) + \frac{1}{1+\lambda_i} \left[\phi_i^j(m) - (1+\lambda_i)\, C_i^j(m) \right] \right\}$ *for* $i = \alpha, \beta$, *and*

4) $\exists\, m^{-i} \in \arg\max \left\{ G(m; C_\alpha^j, C_\beta^j) \right\}$ *with* $C_i^j(m^{-i}) = 0$.

Condition 1 simply states the implicit assumption that lobbies cannot demand contributions from the government. Condition 2 states that the government chooses a policy that maximizes its payoff given the contribution schedules of the lobbies. Condition 3 states that in equilibrium, the government chooses a policy that maximizes the weighted joint payoff of the government and lobby i, the weight of the lobby depending on its efficiency. A violation of this condition at $\widetilde{m}$ would imply that lobby i could induce government to choose $\widetilde{m}$ instead of m^j by changing its contribution schedule in such a way that both lobby i and the government were strictly better off. Condition 4 in turn states that there must exist policy m^{-i} that gives the government the same payoff as the equilibrium policy and induces a zero contribution from lobby i. If not, lobby i could reduce its equilibrium contribution without inducing the government to deviate from the equilibrium policy.

As is customary in the political applications of the common agency framework, we assume that the contribution schedules of the lobbies are truthful. This means that the change in contribution induced by a given policy change always correctly reflects the policy preferences of the lobbies. Conditions of Lemma 1 imply that the equilibrium policy for $j = d, e$ is implicitly defined by

$$\frac{\partial \phi_i^j(m^j)}{\partial m} = (1+\lambda_i)\, \frac{\partial C_i^j(m^j)}{\partial m} \text{ for } i = \alpha, \beta. \tag{11}$$

Hence, in the equilibrium, the marginal change in total lobbying costs of lobby i induced by a small change in conservation policy equals the marginal change in the welfare of lobby i. The cost of lobbying essentially dampens the effect that a given policy change has on the contribution of lobby i. Thus, when the lobbying cost increases, the equilibrium contribution of the lobby becomes less responsive to changes in welfare.

Plugging equation (11) into the first-order condition of the government derived from condition 2 of Lemma 1 gives

$$\left[\frac{1}{(1+\lambda_\alpha)}+a\right]\frac{\partial\phi_\alpha^j(m^j)}{\partial m}+\left[\frac{1}{\left(1+\lambda_\beta\right)}+a\right]\frac{\partial\phi_\beta^j(m^j)}{\partial m} + a\frac{\partial\phi_{1-\alpha-\beta}^j(m^j)}{\partial m}=0. \tag{12}$$

Thus, the policy that maximizes the payoff of the government according to condition 2 of Lemma 1 also maximizes

$$\Omega^j(m)= \left[\frac{1}{(1+\lambda_\alpha)}+a\right]\phi_\alpha^j(m)+\left[\frac{1}{\left(1+\lambda_\beta\right)}+a\right]\phi_\beta^j(m)+a\phi_{1-\alpha-\beta}^j(m). \tag{13}$$

The discussion above implies that the outcome of lobbying is efficient in the sense that it maximizes the weighed sum of the payoff of the government and the two active lobbies.[15] However, to the extent that not all consumers who are influenced by conservation decisions engage in lobbying activities, the politically determined conservation level will be inefficient from the point of view of the society as a whole.

4. Conservation as a Result of the Political Process

In this section, we analyze the relationship of the political equilibrium and the socially optimal conservation level and determine factors that influence the equilibrium outcome. We first analyze the relationship of the political equilibria when the wood product is exported and when it is consumed domestically and study how the market power of the wood processing industry influences the political equilibria. A comparison of the two political equilibria determined by (12) shows that

Proposition 1 *When the final product of the wood processing industry is exported, the politically determined conservation policy is stricter than when the wood product is destined for domestic consumption.*

Proof. By using equations (4)–(7) it follows that for $j=d$

$$\Omega_m^d(m)=\left[\frac{1}{1+\lambda_\alpha}+a\right]\pi_m\left(p,q(m),w\right)+\varphi\left[cs_m(m)+g'(m)\right]$$

where $\varphi=\left[a+\frac{\alpha}{(1+\lambda_\alpha)}+\frac{\beta}{\left(1+\lambda_\beta\right)}\right]$. For $j=e$, (12) can be rewritten as

$$\left[\frac{1}{1+\lambda_\alpha}+a\right]\pi_m\left(p,q(m^e),w\right)+\varphi g'(m^e)=0.$$

Evaluating $\Omega_m^d(m)$ at $m=m^e$ then yields $\Omega_m^d(m^e)=\varphi cs_m(m^e)<0$. ■

This result is driven by the effects of conservation on the consumer surplus. As the wood processing industry is non-competitive, the optimal output is generally lower than the output that would prevail in a competitive market. A strict conservation policy further reduces the supply of the wood processing industry. This reduces consumer surplus and affects all the consumers when the wood product is consumed domestically.[16] In contrast, when the wood product is exported, the negative effects of conservation are restricted to the owners of the wood processing industry. The costs of forest conservation are therefore partly borne by foreign consumers.

In order to study the effect of the market power on the politically determined forest conservation policy, for tractability, we concentrate on a situation where the demand elasticity is constant. This allows us to use the degree of the demand elasticity as a measure of the market power of the monopoly. Let $y_\varepsilon(m)$ denote the marginal influence of a change in the demand elasticity on the profit maximizing output. We then have that

Lemma 2 *When $y_\varepsilon(m) > 0$ and the wood product is exported, the equilibrium conservation level is increasing in the market power of the exporting monopoly. If the wood product is destined for the domestic market, the effect of market power on the equilibrium policy is ambiguous.*

Proof. See Appendix B. ■

If the optimal output level is increasing in the demand elasticity, the negative effect of conservation on profit is aggravated by loss of market power. When the wood product is exported, profit is the only channel through which market power influences the effect of conservation decisions on aggregate welfare. Consequently, if the monopoly loses market power, overall social costs of forest conservation increase while the benefits remain unchanged. Thus, a deterioration in the market conditions of the exporting monopoly makes the government reluctant towards forest conservation. Hence, the conservation level is lower when the domestic wood processing industry loses market power in the export market.

When the product of the wood processing industry is destined for domestic markets, a change in the demand conditions affects the equilibrium policy through two channels. As in the case of exported wood product, the negative effect of conservation on profits is aggravated by loss of market power. In addition, changes in the market power of the industry influence the effect of conservation on the consumer surplus. However, this effect remains ambiguous. Therefore, when the wood product is consumed domestically, the influence of changes in market power on the political equilibrium is ambiguous.

Consider then how the politically determined conservation policy may be distorted from the social optimum. Here, lobbying may lead the conservation policy to be inefficient for three distinct reasons: because the wood processing industry has market power, because the two lobbies are not equally efficient in lobbying, and because not all who benefit from forest conservation take part in lobbying.[17]

Proposition 2 *When the wood product is exported or consumed domestically*

i) if the industrial lobby is more efficient than the environmental lobby, the politically determined conservation level is insufficient from the social point of view.

ii) if the environmental lobby is more efficient than the industrial lobby, depending on how large the group of politically passive consumers is relative to the group of environmentalists, conservation may be insufficient or excessive. Conservation is more likely to be insufficient when the group of politically passive consumers is large.

Proof. See Appendix B. ■

When the wood product is exported, the only group that is adversely affected by an increase in the conservation level is the owners of the monopoly. All the other consumers benefit from stricter conservation policies. When the wood product is consumed domestically, the conservation decision influences all consumers through the consumer price. In both cases, if the industrial lobby is at least as efficient as the environmental lobby, the politically determined conservation level is insufficient relative to the socially optimal conservation level.

However, the conservation level may remain too low even if the environmental lobby is more efficient than the industrial lobby. This happens especially if the group of politically passive consumers is large. The reason is that the passive consumers benefit from the lobbying effort of the environmental lobby, but the objective of the group is to maximize the welfare of its members only. This means that the environmental lobby does not take into account the benefits of its lobbying effort for the other groups in the society. The politically determined conservation will be excessive only if the environmental lobby is sufficiently more efficient than the industrial lobby. This happens if the difference in lobbying efficiencies is substantial relative to the ratio of the population share of the politically passive consumers and the politically active environmentalists.

Making lobbying prohibitively costly for the two lobbies would guarantee that policy design reflects the social optimum. However, without precise knowledge of the factors influencing the efficiency of lobbying, this policy could lead to substantial inefficiencies. As mentioned earlier, the nature of the lobby is likely to affect the efficiency. For instance, it is conceivable that a lobby presenting a concentrated industry is able to control free riding better than a large environmental group with a dispersed membership. The efficiency is also likely to depend on the institutional environment. For instance, it is reasonable to assume that the efficiency of lobbying varies from one country to another and even from one policy issue to another and that the preferences of the government in power also influence the relative efficiency of the lobbies.

5. Conclusions

This paper has used a political economy approach to analyze the design of forest policy. More specifically, we have studied how forest conservation policy is determined when the government is influenced by an industrial and environmental lobby. Policy formation has been modelled as a common agency game.

Comparison of the political equilibria shows that an exporting monopoly faces a stricter conservation policy than a monopoly whose production is destined for the domestic market. This is because when the wood product is exported, part of the costs of conservation are borne by foreign consumers. The more elastic the demand of the wood product abroad, the more severely the wood processing industry is affected by an increased conservation requirement. Hence, a change in the international environment that causes the domestic industry to face more elastic demand abroad limits the willingness of the government to engage in forest conservation.

When the wood product is exported, conservation decisions influence the profits of the wood processing industry but affect all consumers only through the benefit derived from non-timber services of forests. Hence, the environmentalists and the politically passive consumers always benefit from stricter conservation policies. When the wood product is destined for the domestic markets, an increased conservation requirement affects the welfare of all the domestic consumers through a higher price of consumption. In both cases, when the industrial lobby is more efficient than the environmental lobby, conservation policy in the political equilibrium is insufficient compared to the socially optimal conservation level. But conservation may be insufficient from the social point of view even if the environmental lobby is more efficient than the industrial lobby. This is likely to happen if the group of the politically passive consumers is large.

Here we have considered only one policy instrument. One interesting topic for future research would be to study whether some policy instruments attract more pressure from special interests than others. If so, an important normative question is, should the instrument set available to the government be restricted in some way in order to ensure that policies reflecting the social optimum are chosen. Another avenue for further research concerns the issue of who in the end benefits from conservation of forests. Addressing this question will then lead to questions like is it socially beneficial to support the work of public good pressure groups from public funds.

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Notes

1. See e.g., Koskela and Ollikainen (1999).
2. For a discussion on forestry conflicts, see Palo and Uusivuori (1999) and references therein.
3. For a discussion on the research of special interest politics, see Persson and Tabellini (2000) and Grossman and Helpman (2001).
4. See e.g., Koskela and Ollikainen (1998) and Bergman and Brännlund (1995).
5. In order to clearly identify the effects of forest conservation on consumer surplus, we concentrate on the situation where the domestic production is not competing with imports.
6. See Appendix A for details.
7. Partial derivatives are denoted by $\frac{\partial \Omega}{\partial x}$, by Ω_x, or, in the case of functions of a single variable, by primes.
8. As only two groups participate to the political process, it is conceivable that they could reach an agreement on the appropriate level of conservation through negotiation. Since typically the most visible way of influencing conservation decisions is the direct influence on government decision making, we concentrate on that situation.
9. It is conceivable that the monopoly lobby is able to control free riding of its members fairly efficiently. The same may not be true for large environmental organizations. See Olson (1965), Stigler (1971), and Peltzman (1976) for seminal contributions on this issue. For endogenous formation of lobby groups in the common agency setting, see Mitra (1999).
10. As usual, $u' > 0$, $u'' < 0$, $g' > 0$, and $g'' < 0$. Both $u\,(y)$ and $g\,(m)$ satisfy Inada conditions.
11. An alternative assumption would be that only consumers belonging to the environmental lobby benefit from conservation, for instance, because they value the recreational opportunities created by forests.
12. See e.g., Grossman and Helpman (1994) and Goldberg and Maggi (1999) for discussion.
13. For instance, in Fredriksson (1997) environmental policy under lobbying would coincide with the socially optimal policy, if the tax revenue was returned to the polluting industry instead of being distributed to all consumers.
14. A more complete analysis would then incorporate elections where the identity of the government is determined, see Grossman and Helpman (1996). Here, as is typical in the literature, the transaction cost is exogenous. Genuine incorporation of the features influencing efficiency would call for endogenization of this cost, see Faure-Grimaud et al. (1999).
15. For equations (12) to characterize maxima, second-order conditions must hold. If $\Omega^j_{mm}\,(m) \leq 0$, $\Omega^j_m(m^j) = 0$ determines a unique equilibrium conservation level. Throughout the analysis of the following section, we will assume that the second-order conditions are satisfied.
16. See e.g., Barnett (1980) on the regulation of a polluting monopoly.
17. A fourth possible source inefficiency would be the existence of tax revenue as it creates an additional incentive for lobbying.

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Appendix A

Lagrangian of the constrained optimization problem of the monopolist is

$$L\left(k, n, \lambda\right) = p\left(f\left(k, n\right)\right) f\left(k, n\right) + \left(\overline{k} - k\right) \overline{q} - wn + \lambda \left(\overline{k} - k\right).$$

Using $\varepsilon = -\frac{df(k,n)}{dp(\cdot)}\frac{p(\cdot)}{f(k,n)}$ the associated first-order conditions are

$$L_k(k,n,\lambda) = 0 \Leftrightarrow \left(1-\frac{1}{\varepsilon}\right) pf_k(k,n) = \overline{q}+\lambda, \tag{A1}$$

$$L_n(k,n,\lambda) = 0 \Leftrightarrow \left(1-\frac{1}{\varepsilon}\right) pf_n(k,n) = w, \text{ and} \tag{A2}$$

$$L_\lambda(k,n,\lambda) = 0 \Leftrightarrow \overline{k}-k = 0. \tag{A3}$$

Define $q = \overline{q}+\lambda$. The first-order conditions establish a maximum, since the bordered Hessian matrix

$$H = \begin{bmatrix} 0 & -1 & 0 \\ -1 & \pi_{kk} & \pi_{kn} \\ 0 & \pi_{nk} & \pi_{nn} \end{bmatrix}$$

is negative semi-definite. Totally differentiating (A1)–(A3) with respect to λ and $\overline{k}$ yields

$$\begin{bmatrix} \frac{\partial^2 L}{\partial\lambda^2} & \frac{\partial^2 L}{\partial\lambda\partial k} & \frac{\partial^2 L}{\partial\lambda\partial n} \\ \frac{\partial^2 L}{\partial k\partial\lambda} & \frac{\partial^2 L}{\partial k^2} & \frac{\partial^2 L}{\partial k\partial n} \\ \frac{\partial^2 L}{\partial n\partial\lambda} & \frac{\partial^2 L}{\partial k\partial n} & \frac{\partial^2 L}{\partial n^2} \end{bmatrix} \begin{bmatrix} \frac{d\lambda}{d\overline{k}} \\ \frac{dk}{d\overline{k}} \\ \frac{dn}{d\overline{k}} \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}.$$

Thus when $\overline{k} = k$,

$$\frac{dq}{d\overline{k}} = \frac{d\lambda}{d\overline{k}} = \frac{-\left[\pi_{kk}\pi_{nn}-\pi_{kn}\pi_{nk}\right]}{|H|} < 0 \text{ and } \frac{d^2q}{d\overline{k}^2} = 0.$$

Appendix B

Proof of Lemma 2. When $\overline{k}-k=0$, the profit function of the monopoly is

$$\pi(p, q(m), w) = \max_y \{p(y)\,y - c(q(m), w)y\}, \tag{B1}$$

and hence the optimal output level is determined by

$$p(y) + p'(y)\,y - c(q(m), w) = 0. \tag{B2}$$

Using $p = y^{-\frac{1}{\varepsilon}}$ where $\varepsilon = -\frac{dy}{dp}\frac{p}{y}$ gives $y(m) = \left[\frac{c(q(m),w)}{1-\frac{1}{\varepsilon}}\right]^{-\varepsilon}$ and

$$y_m(m) = -\varepsilon y(m)\frac{c_m(q(m), w)}{c(q(m), w)} < 0. \tag{B3}$$

If ε is constant

$$\begin{aligned} y_\varepsilon(m) &= y(m)\left[-\ln\left[\frac{c(q(m), w)}{1-\frac{1}{\varepsilon}}\right] + \frac{1}{\varepsilon-1}\right] \\ &= y(m)\left[-\ln\left[y(m)^{-\frac{1}{\varepsilon}}\right] + \frac{1}{\varepsilon-1}\right]. \end{aligned}$$

When $y(m) > 1$ the optimal output is increasing in the demand elasticity. Totally differentiating first-order condition (12) yields

$$\frac{dm^j}{d\varepsilon} = -\frac{\Omega^j_{m\varepsilon}\left(m^j\right)}{\Omega^j_{mm}\left(m^j\right)},$$

where $\Omega^j_{mm}\left(m^j\right) < 0$. For $j = e$

$$\Omega^e_{m\varepsilon}\left(m^e\right) = -\left(\frac{1}{1+\lambda_\alpha} + a\right) y_\varepsilon(m)\, c_m\left(q\left(m^e\right), w\right).$$

If $y_\varepsilon(m) > 0$ it directly follows that $\Omega^e_{m\varepsilon}(m^e) < 0$. As a result, $\frac{dm^e}{d\varepsilon} < 0$. For $j = d$

$$\Omega^d_{m\varepsilon}(m^d) = \left(a + \frac{1}{1+\lambda_\alpha}\right)\pi_{m\varepsilon}\left(p, q\left(m^d\right), w\right) + \left(a + \frac{\alpha}{1+\lambda_\alpha} + \frac{\beta}{1+\lambda_\beta}\right) cs_{m\varepsilon}\left(m^d\right).$$

Using $p = y^{-\frac{1}{\varepsilon}}$ gives that

$$cs_m(m) = -\frac{\varepsilon c_m(q(m), w)}{\varepsilon - 1} y(m)$$

and hence

$$cs_{m\varepsilon}(m) = \frac{c_m(q(m), w)}{(\varepsilon-1)^2} y(m) - \frac{\varepsilon c_m(q(m), w)}{\varepsilon - 1} y_\varepsilon(m).$$

When $y_\varepsilon(m) > 0$, the effect of market power on consumer surplus remains ambiguous. ■

Proof of Proposition 2. Solving (7) and (9) for $\frac{\partial\phi^j_\alpha(m)}{\partial m}$ and plugging into (12) gives

$$\left(\frac{1}{1+\lambda_\alpha} + a\right)\frac{\partial\phi^j(m^j)}{\partial m} + \frac{1}{1+\lambda_\alpha}\left[\frac{\lambda_\beta - \lambda_\alpha}{1+\lambda_\beta}\frac{\partial\phi^j_\beta(m^j)}{\partial m} + \frac{\partial\phi^j_{1-\alpha-\beta}(m^j)}{\partial m}\right] = 0.$$

Hence, for $j = d$, the marginal aggregate welfare evaluated at $m = m^d$ is

$$\begin{aligned}\frac{\partial\phi^d(m^d)}{\partial m} &= \frac{1}{(1 + a(1+\lambda_\alpha))}\left[\frac{\lambda_\beta - \lambda_\alpha}{\left(1+\lambda_\beta\right)}\frac{\partial\phi^d_\beta(m^d)}{\partial m} + \frac{\partial\phi^d_{1-\alpha-\beta}(m^d)}{\partial m}\right] \\ &= \frac{\left[(1-\alpha-\beta)\left(1+\lambda_\beta\right) + \beta\left(\lambda_\beta - \lambda_\alpha\right)\right]\left[g'\left(m^d\right) + cs_m\left(m^d\right)\right]}{\left(1+\lambda_\beta\right)(1 + a(1+\lambda_\alpha))}.\end{aligned}$$

In the same manner, for $j = e$,

$$\frac{\partial\phi^e(m^e)}{\partial m} = \frac{\left[(1-\alpha-\beta)\left(1+\lambda_\beta\right) + \beta\left(\lambda_\beta - \lambda_\alpha\right)\right] g'(m^e)}{(1 + (1+\lambda_\alpha)a)\left(1+\lambda_\beta\right)}$$

Since $g'(m) > 0$ and $g'\left(m^d\right) + cs_m\left(m^d\right) > 0$, a sufficient condition for $\frac{\partial\phi^d(m^d)}{\partial m} > 0$ and $\frac{\partial\phi^e(m^e)}{\partial m} > 0$ is $\lambda_\alpha < \lambda_\beta$. A necessary condition for $\frac{\partial\phi^d(m^d)}{\partial m} < 0$ and $\frac{\partial\phi^e(m^e)}{\partial m} < 0$ in turn is that $\lambda_\alpha - \lambda_\beta > \frac{1-\alpha-\beta}{\beta}\left(1+\lambda_\beta\right)$. ■